Desca Affajry Rais

Mathematics Undergraduate · Researcher · Educator

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ACADEMIC & PROFESSIONAL PROFILE

I am a final-year Mathematics undergraduate at Universitas Diponegoro with a GPA of 3.92/4.00. My primary interests lie in pure mathematics, particularly real and complex analysis, graph theory, abstract algebra, topology, and combinatorics. Alongside my academic work, I contribute to undergraduate mathematics education, olympiad coaching, assessment development, data analysis, and computational modelling.

EDUCATION

Sep 2023 — Present
ONGOING

Universitas Diponegoro
B.Sc. in Mathematics · GPA 3.92 / 4.00

Chair of UKM KOMIPA UNDIP (2025); delivered tutorials in Real Analysis II; contributed to Himpunan Mahasiswa Matematika Universitas Diponegoro; selected as an ON-MIPA delegate and recognised in mathematics competitions.

Jun 2020 — Mar 2023
COMPLETED

SMA Negeri 1 Sulang
Mathematics and Natural Sciences · Final Grade 86.55 / 100.00

Received awards in regional and national mathematics competitions, represented SMA Negeri 1 Sulang in Olimpiade Sains Nasional (OSN) for Informatics, and provided peer mathematics mentoring.

RESEARCH & ACADEMIC WORKS

2026
DRAFT

Distance Spectra of Corona Graphs of Complete Graphs and Paths via Invariant Subspaces
Desca Affajry Rais · Suryoto, S.Si., M.Si.

Studies the distance spectrum of $K_n \odot P_m$ using block matrices, Kronecker products, and invariant subspaces.

2026
DRAFT

Connectivity of Int-Graphic Topology on Directed Graphs
Desca Affajry Rais · YD Sumanto · Solikhin

Characterises connectivity in finite int-graphic Alexandroff spaces using minimal open neighbourhoods and the preorder induced by out-neighbourhood inclusion.

2026
PRESENTED

Modular Total Edge Irregular Labeling of Corona Product of Cycle C_4 with Path and Null Graphs
The 16th ISNPINSA · Oral Presentation

Developed explicit labelling constructions and established exact values of modular total edge irregularity strength.

2026
COMPLETED

Inequality Problem Solving Book
Desca Affajry Rais · Drs. YD Sumanto, M.Si. · First Edition · 174 pages

A 174-page mathematical manuscript combining foundational inequality theory, rigorous proofs, worked examples, and progressively challenging exercises.

PROFESSIONAL EXPERIENCE

Aug 2026 — Present
ONGOING

University Mathematics Tutor
Tutorkuliah.id

Provided advanced undergraduate mathematics tuition emphasising conceptual clarity and rigorous problem solving.

Jan 2026 — Present
ONGOING

Mathematics Olympiad Coach
KOMIPA UNDIP

Delivered university-level olympiad coaching in Real Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, and Combinatorics.

Apr 2026 — Jun 2026
COMPLETED

Teaching Assistant
Asosiasi Matematika Terapan UNDIP

Facilitated Real Analysis tutorials, proof-based problem-solving sessions, and individual academic guidance.

Dec 2025 — Mar 2026
COMPLETED

Mathematics Item Writer
SainsIn

Developed original quantitative-reasoning assessments, fully worked solutions, instructional videos, and quality-assurance materials.

Nov 2025 — Jan 2026
COMPLETED

Data Analyst
Emergencyy Call

Prepared analytical datasets, executed SQL queries, produced Excel reports and KPI dashboards, and documented data workflows.

Nov 2025 — Dec 2025
COMPLETED

Mathematics Item Writer
SNBT.ID

Developed demanding Quantitative Knowledge and Mathematical Reasoning assessments with logically structured solutions.

May 2025
COMPLETED

Project-Based Virtual Intern · Data Scientist
ID/X Partners × Rakamin Academy

Developed and evaluated a credit-risk modelling workflow using Logistic Regression and Random Forest.

Mar 2025
COMPLETED

Project-Based Virtual Intern · Big Data Analyst
PT Kimia Farma Tbk × Rakamin Academy

Constructed an analytical dataset in BigQuery and a multidimensional performance dashboard in Looker Studio.

LEADERSHIP & ORGANIZATIONS

Jan 2025 — Dec 2025
COMPLETED

Chairman
UKM KOMIPA UNDIP

Oversaw strategic planning, daily operations, olympiad mentoring, ON-MIPA delegations, and external representation.

Jan 2025 — Dec 2025
COMPLETED

Senior Staff, Education and Reasoning Department
Himpunan Mahasiswa Matematika Universitas Diponegoro

Delivered mathematics tutorials and developed instructional content and learning resources.

Feb 2024 — Jan 2025
COMPLETED

Junior Staff, Education and Reasoning Department
Himpunan Mahasiswa Matematika Universitas Diponegoro

Coordinated the Academic Delegation programme, supported mentoring for more than 100 students, and received Best Junior Staff and Best Department Staff recognition.

Oct 2023 — Dec 2024
COMPLETED

Staff, Networking and Public Relations Department
KOMIPA UNDIP

Supported ON-MIPA selection, served as Mathematics Liaison Officer, and chaired UKM Roadshow 2024.

2020 — 2023
COMPLETED

Moderator, Mentor, and Board of Governor
Brainly

Progressed through successive moderation and mentoring roles, including the supervision of 29 apprentice moderators.

2020 — 2022
COMPLETED

Organizational Leadership
Pelajar Islam Indonesia (PII), Rembang

Served successively as Head of Organisational Development and General Secretary of the Rembang Regional Board.

COMMITTEE & VOLUNTEER EXPERIENCE

Facilitator Division Coordinator
27th Undip's Mathematics Competition · Jan 2025 — Feb 2026

Curriculum Division Staff
LKMM-TD FSM 2025 · Jan — Oct 2025

Event Division Staff
Introduction, Gathering, & Adaptation Mathematics 2025 · Mar — Oct 2025

Event Division Staff
Industrial Visit Mathematics 2025 · Mar — Aug 2025

Equipment Division Staff
Math Action for Social and Education 2025 · Mar — Jun 2025

Public Relations Division Staff
Leadership Training FSM 2025 · Mar — Apr 2025

Event Division Coordinator
26th Undip's Mathematics Competition · Mar 2024 — Feb 2025

Chief Executive
UKM Roadshow 2024 · Jun — Sep 2024

Volunteer Teacher
Mathematics Clinic 2024 · Aug 2024

Equipment Division Coordinator
ON-MIPA UNDIP Selection 2024 · Oct 2023 — Feb 2024

Mathematics Liaison Officer
ON-MIPA Coaching Center 2024 · Feb 2024

SELECTED PROJECTS

Feb 2026
COMPLETED

Crowd Optimization and Smart Routing Strategy at Dufan Ancol
Operations Research · Mathematical Modeling

Applied M/M/s queueing, Little's Law, greedy optimization, and knapsack concepts; proposed 14,000 daily visitors, a <45-minute waiting-time target, and 75% maximum utilization.

Feb 2026
ONGOING

DMath Learning
Mathematics Education · EdTech

Developing a mathematics learning platform combining structured resources, guided practice, problem banks, olympiad preparation, and supplementary learning tools.

Feb 2026
COMPLETED

Bike Sharing Analytics
Python · EDA · Streamlit

Identified weekday demand peaks, weekend patterns among casual users, and environmental factors associated with bike-sharing demand.

Dec 2025
COMPLETED

Financial Anomaly Detection System
Scikit-Learn · K-Means · Random Forest

Integrated unsupervised pattern discovery with supervised classification to investigate anomalous financial transactions.

May 2025
COMPLETED

Credit Risk Prediction Analysis
SMOTE · Logistic Regression · Random Forest

Analysed 466,285 observations; the reported Random Forest model achieved 99.70% accuracy, 99.94% precision, and 99.46% recall.

Mar 2025
COMPLETED

Kimia Farma Business Performance Analysis
BigQuery · SQL · Looker Studio

Developed a consolidated analytical dataset and an interactive, multidimensional performance dashboard.

Mar–Apr 2025
COMPLETED

Investment Portfolio Optimization
Markowitz · Efficient Frontier · Sharpe Ratio

Examined risk–return trade-offs in portfolios comprising KLBF.JK, HEAL.JK, and KAEF.JK.

SCHOLARSHIPS & TRAINING

Nov 2024 — Present
ONGOING

BSI Scholarship Inspirasi Batch 4
A scholarship programme incorporating academic development, leadership coaching, mentoring, professional workshops, and social-impact activities.

A scholarship programme incorporating academic development, leadership coaching, mentoring, professional workshops, and social-impact activities.

Feb — Jul 2026
COMPLETED

Coding Camp 2026 — Data Scientist
Coding Camp powered by DBS Foundation

Coding Camp powered by DBS Foundation

Mar — Jun 2026
COMPLETED

Microsoft Elevate Training Center
Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Jan — Mar 2026
COMPLETED

IDCamp 2025 — Intermediate Data Scientist
Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

Oct — Dec 2025
COMPLETED

IDCamp 2025 — Beginner Data Scientist
Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

CERTIFICATIONS

Indonesian Language Proficiency Test (UKBI)
Score 724 · Sangat Unggul · 21 Oct 2026 · Valid until 21 Oct 2028

UNDIP SEU English Proficiency Test
EPT 75 · SEU TOEFL Prediction equivalent 601 · Tested 15 Oct 2026 · Issued 22 Oct 2026 · Valid through 15 Oct 2028

AI Engineer Bootcamp

Foundations of Artificial Intelligence

Foundations of SQL

Computer Networking Fundamentals

Data Analyst — Basic Level
ITBOX · Mar 2025

Python Programming Essentials: A Beginner's Guide
Rakamin Academy · Mar 2025

Classical Cryptography for Beginners
Cyber Academy Indonesia · Jul 2024

Introduction to Information Security
Cyber Academy Indonesia · Jul 2024

Student Leadership and Management Training — FSM 2024
BEM FSM UNDIP · Apr 2024

Introduction to Data Science with Python
DQLab · Jan 2024

Python Fundamentals for Data Science
DQLab · Jan 2024

Basic Student Management Skills Training
Sep 2023

Python Programming for Beginners
CODEPOLITAN · Credential Z7R8TDD · Jul 2024 — Jul 2027

SKILLS

Mathematics & Theory
Pure Mathematics · Real Analysis · Complex Analysis · Graph Theory · Abstract Algebra · Topology · Combinatorics · Mathematical Modeling · Formal Proof Writing

Technical & Computing
Python · SQL · LaTeX · Scikit-Learn · Looker Studio · Microsoft Excel · Microsoft Fabric · Microsoft Azure

Data & Analytics
Data Analysis · Exploratory Data Analysis · Machine Learning · Data Visualization · Statistical Analysis · Data Engineering

Academic & Teaching
Research · Academic Writing · Mathematics Olympiad · Teaching · Curriculum Development · Assessment Design · Problem Solving

AWARDS & ACHIEVEMENTS

2nd Place · Calculus Competition · Universitas Halu Oleo
Oct 2026

Silver Medalist · Isaac Newton Competition 2026 · OLIMNUS × UIN Sunan Kalijaga
Jul 2026

Silver Medalist · Future Science Olympiad · POSI
Jun 2026

Bronze Medalist · Mathematical Analysis and Geometry Day XVII · Institut Teknologi Bandung
May 2026

Gold Medalist · Cendekia Natural Science Competition
Apr 2026

1st Place · ON-MIPA UNDIP Selection 2026
Apr 2026

1st Place · Indonesia Science Competition (KSI)
Mar 2026

Gold Award · KTOM March 2026
Mar 2026

1st Place · Indonesia Science Competition (InSC)
Feb 2026

4th Place · Honorable Mention 1 · MATICS Mathematics Olympiad · Hasanuddin University
Feb 2026

Silver Award · KTOM February 2026
Feb 2026

Bronze Medalist · POSI Science Competition
Jan 2026

2nd Place · Indonesian Youth Science Olympiad #6
Jan 2026

2nd Place · National Science League #2
Jan 2026

Silver Medalist · Einstein Science Competition 2026
Jan 2026

Gold Award · KTOM December 2025
Dec 2025

Special Honour Participant · International Youth Math Challenge
Dec 2025

1st Place · LOSEMA X
Nov 2025

1st Place · Olimnus Science 2025
Nov 2025

Silver Medalist · Isaac Newton Competition 2025
Jul 2025

3rd Place · ASOE 4.0
May 2025

4th Place · ON-MIPA UNDIP Selection 2025
Mar 2025

Finalist · MATHLICIOUS 2022
Nov 2022

1st Place · ALHAMBRA Mathematics Competition
Feb 2022

Bronze Medalist · Insight Arjuna Competition
Oct 2021

Gold Medalist · Dirgantara Science Competition
Oct 2021

Silver Medalist · Indonesian Youth Science Competition
Oct 2021

Bronze Medalist · Bimage Competition Season 2
Sep 2021

Silver Medalist · POSI Science Competition
Sep 2021

Top 10 · National Calculus Olympiad Nusantara
Aug 2021